

The empirical recurrence for OEIS A295916: recovering a conjecture that is not written down

Adrian Perez Fontelles
Independent researcher

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Abstract

OEIS A295916 records “Empirical recurrence of order 92”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 183$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 92, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 92 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A295916 is “Number of nX6 0..1 arrays with each 1 adjacent to 0 or 3 king-move neighboring 1s.”. It has offset 1 and begins

21, 129, 993, 7604, 56955, 431844, 3261576, 24650278, ...

The entry states:

Empirical recurrence of order 92 (see link above)

As of the “Last modified” line on the live entry (Nov 29 2017 18:19:13, revision 6) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 4160$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 183$ of the 4160, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 183$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 366$ exact terms of the model returns order 92 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{92} c_i a(n-i),$$

c_1	3	c_{24}	412438858737	c_{47}	−4580394703995726	c_{70}	−20331040896657
c_2	31	c_{25}	−451147269589	c_{48}	−1350254352385509	c_{71}	−11009984342291
c_3	351	c_{26}	−2398968634713	c_{49}	2469692051425512	c_{72}	4985740671561
c_4	−1013	c_{27}	−2082908557359	c_{50}	4331058262389879	c_{73}	5893755434579
c_5	−9714	c_{28}	3218198907216	c_{51}	488248301681521	c_{74}	626211272839
c_6	−46311	c_{29}	11316894882221	c_{52}	−2691744824773241	c_{75}	−1602944545910
c_7	123075	c_{30}	5914053584393	c_{53}	−2100751814256350	c_{76}	−1015507843531
c_8	1061152	c_{31}	−11209025152309	c_{54}	1909405156088453	c_{77}	122018013629
c_9	3006522	c_{32}	−27734474362454	c_{55}	2172184779287738	c_{78}	300988858913
c_{10}	−6931754	c_{33}	2168082344059	c_{56}	−850694172432111	c_{79}	63645862103
c_{11}	−51973022	c_{34}	14041696912282	c_{57}	−2545252440853188	c_{80}	−38260317800
c_{12}	−101735690	c_{35}	−29227435020883	c_{58}	35955806128465	c_{81}	−22096516058
c_{13}	190480606	c_{36}	−92820941151241	c_{59}	1581964901177897	c_{82}	1771216035
c_{14}	1190312371	c_{37}	68319843402047	c_{60}	661517028214403	c_{83}	2952111733
c_{15}	1674703534	c_{38}	450983043138428	c_{61}	−755090462100923	c_{84}	214877950
c_{16}	−2434608509	c_{39}	343034220471349	c_{62}	−478612743863736	c_{85}	−165175712
c_{17}	−11566320145	c_{40}	−528656300364053	c_{63}	249918549862338	c_{86}	−74718908
c_{18}	−9622078598	c_{41}	−1624373533180706	c_{64}	255353099530895	c_{87}	−551128
c_{19}	12015446926	c_{42}	−649299358435102	c_{65}	−65219984913105	c_{88}	10778928
c_{20}	31467296828	c_{43}	1654817610782703	c_{66}	−136076690405487	c_{89}	617632
c_{21}	−21549089037	c_{44}	3356039669580101	c_{67}	12498745527913	c_{90}	−743040
c_{22}	20218167870	c_{45}	975550962305896	c_{68}	58382503762937	c_{91}	−29952
c_{23}	224043625216	c_{46}	−2550823643447448	c_{69}	7929310417148	c_{92}	19968

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 92, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $92 + 4$ terms removed returns order 92 as well, so $\deg q_{\min} = 92 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 19 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once

more on the sequence with its first $92 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 92 that this sequence satisfies — and that family turns out to have exactly one member.

References

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